

Big Data

Apuntes

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2026-05-12

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1. Frameworks and tools

Specialized tools and frameworks are required for big data analysis when:

1. The volume of data involved is so large that is difficult to store, process and analyze data on a single machine.
2. The velocity of data is very high and the data needs to be analyzed in real-time.
3. There is a variety of data involved, which can be structured, unstructured or semi-structured, and is collected from multiple data sources.
4. Various types of analytics need to be performed to extract value from the data such as descriptive, diagnostic, predictive and prescriptive analytics.

Big Data tools and frameworks have distributed and parallel processing architectures and can leverage the storage and computational resources of a large cluster of machines.

Analytics involve several steps starting from data cleansing, data munging (or wrangling), data processing and visualization.

Specialized tools and framework are required to ingest the data from different sources into the big data analytics backend.

Based on the analysis requirements (batch or real-time), and type of analysis to be performed (descriptive, diagnostic, predictive, or prescriptive) specialized frameworks are used.

Big data analytics is enabled by several technologies such as:

- cloud computing.
- distributed and parallel processing frameworks.
- non-relational databases.
- in-memory computing.
- etc.

2. Characteristics of Big Data

2.1. Volume

Big data is form of data whose volume is so large that it would not fit on a single machine therefore specialized tools and frameworks are required to store process and analyze such data.

2.2. Velocity

- Velocity of data refers to how fast the data is generated and the primary **reason for the exponential growth of data**.
- Results in the volume of data accumulated to become very large, in short span of time.
- Some application can have strict deadlines for data analysis and the data needs to be analyzed in real-time.
- Specialized tools are required to ingest such high velocity data into the big data infrastructure and analyze the data.

2.3. Variety

Flexible enough to handle such Variety on the forms of the data:

- structured
- unstructured
- semi-structured
- text data
- image
- audio
- video
- sensor data

2.4. Veracity

Veracity refers to how accurate is the data.

- To extract value from the data, the data needs to be cleaned to remove noise.
- Cleansing of data is important so that incorrect and faulty data can be filtered out.

2.5. Value

Value of data refers to the usefulness of data for the intended purpose. The end goal of any big data analytics system is to extract value from the data.

2.6. Data collection

Before the data can be analyzed, the data must be collected and ingested into a big data stack.

The choice of tools and frameworks for data collection depends on the source of data and the type of data being ingested.

2.7. Data preparation

Data can often be dirty and can have various issues that must be resolved before the data can be preprocessed, such as:

- corrupt records.
- missing values.
- duplicates.
- inconsistent abbreviations.
- inconsistent units.
- typos.
- incorrect spellings.
- incorrect formatting.

involves various tasks such as:

- data cleaning.
- data wrangling or munging.
- de-duplication.
- normalization.
- sampling.
- filtering.

Data cleaning detects and resolves issues such as:

- corrupt records.
- records with missing values.
- records with bad formatting.
- for instance.

Data wrangling or munging deals with transforming the data from one raw format to another (Some file may be using comma as the field separator, other may be using tab as the field separator).

Normalization is required when data from different sources uses different units or scales or have different abbreviations for the same thing.

Filtering and sampling may be useful when we want to process only the data that meets certain rules.

2.8. Analysis Types

Analysis types and the algorithms.

2.9. Analysis Modes

To determine the analysis mode, which can be either batch, real-time, or interactive.

The choice of the mode depends on the requirements of the application.

If your application demands results to be updated after short intervals of time (say every few seconds), then real-time analysis mode is chosen.

If your application only requires the results to be generated and updated on larger timescales (say daily or monthly), then batch mode can be used.

If your application demands flexibility to query data on demand, then the interactive mode is useful.

Once you make a choice of the analysis type and the analysis mode, you can **determine the processing pattern** that can be used.

The choice of the analysis type, analysis mode, and the data processing pattern can help you in shortlisting the right tools and frameworks for data analysis.

2.10. Visualizations

The choice of the visualization tools, serving databases and web frameworks is driven by the requirements of the application.

Static visualization are used when you have the analysis results stored in a serving database and you simply want to display the results.

If your application demands the results to **update regularly**, then you would require **dynamic** visualizations.

If you want your application to accept inputs from the user and display the results, then you would require **interactive** visualizations.

2.11. Raw Data Sources

Before the data is processed and analyzed, it must be captured from the **raw data sources** into the big data systems and frameworks.